

Global solutions for the interpolated stochastic heat and wave equation with a super-linear diffusion term

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Abstract

In this paper we prove the existence of a global solution to the interpolated stochastic heat and wave equation driven by a space-time white noise and with a super-linear non-linear diffusion term.

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1 Introduction

In this chapter, we study the interpolated stochastic heat and wave equation (ISHWE) on the whole space $\mathbb{R}^d$,

$$\begin{cases} \left(\partial_t^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2} \right) u(t, x) = I_t^\gamma \left[\sigma(u(t, x)) \dot{W}(t, x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = u_0(x) & \beta \in (0, 1] \\ u(0, \cdot) = u_0(x), \quad \partial_t u(0, \cdot) = v_0(x) & \beta \in (1, 2), \end{cases} \quad (1.1)$$

E:SupLinSPD

where the fractional differential operators ∂_t^β , $(-\Delta)^{\alpha/2}$ and I_t^γ respectively denote the *Caputo derivative*, the *fractional Laplacian* and the *Riemann-Liouville fractional integral operator*. The noise, $\dot{W}$ is taken to be space-time white and the initial data, u_0 and v_0 , are assumed to be

Borel measurable functions which are Bounded on $\mathbb{R}^d$. The nonlinearity, σ , is *locally Lipschitz* in the sense that the following asymptotic relation holds as $|x|, |z| \rightarrow \infty$:

$$|\sigma(x) - \sigma(z)| \leq \sigma_2 |x - z| [\ln_+(|x - z|)]^\delta, \quad (1.2)$$

where $\sigma_2, \delta > 0$ and $\ln_+(z) = \ln(z \vee e)$ for $z > 0$. Further, this implies that as $|x| \rightarrow \infty$,

$$|\sigma(x)| \leq \sigma_1 + \sigma_2 |x| [\ln(|x|)]^\delta, \quad (1.3)$$

where $\sigma_1 = \sigma(0)$.

For any $T > 0$ with $t \in [0, T]$ and $x \in \mathbb{R}^d$, the solution to (1.1) is understood in the *mild* sense as the solution to the following stochastic integral equation:

$$u(t, x) = J_0(t, x) + I(t, x),$$

where

$$\begin{aligned} J_0(t, x) &= \begin{cases} [Z(t, \cdot) * u_0](x) & \beta \in (0, 1] \\ [Z(t, \cdot) * v_0](x) + [Z^*(t, \cdot) * u_0](x) & \beta \in (1, 2) \end{cases} \\ I(t, x) &= \int_0^t \int_{\mathbb{R}^d} Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy). \end{aligned}$$

The stochastic integral above is a *Walsh integral* (e.g. see [Wal86]) and the fundamental solution is a triple, $\{Z, Z^*, Y\}$, and each member of the triple is defined through the Fox-H function, however, one can more compactly define them through their Fourier transforms [CHN19, Theorem 4.1]:

$$\begin{aligned} \mathcal{F}Z(t, \cdot)(\xi) &= t^{\lceil \beta \rceil - 1} E_{\beta, \lceil \beta \rceil} \left(-2^{-1} \nu t^\beta |\xi|^\alpha \right), \\ \mathcal{F}Z^*(t, \cdot)(\xi) &= E_{\beta, 1} \left(-2^{-1} \nu t^\beta |\xi|^\alpha \right), \\ \mathcal{F}Y(t, \cdot)(\xi) &= t^{\beta + \gamma - 1} E_{\beta, \beta + \gamma} \left(-2^{-1} \nu t^\beta |\xi|^\alpha \right), \end{aligned}$$

where $E_{a,b}(z)$ denotes the *two-parameter Mittag-Leffler function* [Pod99]:

$$E_{a,b}(z) := \sum_{k=0}^{\infty} \frac{z^k}{\Gamma(ak + b)}, \quad a > 0, b > 0,$$

and Γ is the standard *Gamma-function*.

We mention that under the special case where $\alpha \in (1, 2)$, $\beta \in (0, 1]$ and $\gamma = 1 - \beta$, that $Z = Z^* = Y$, which is easily seen from their Fourier transforms given above. Because of these equalities, we simply let Y denote the fundamental solution under this situation. Moreover, under this case there is a probabilistic representation for Y which is used by the authors in [FLN19; FN17; MN15; FLO17; MN16]. Indeed, let X_t denote a symmetric α -stable process with density $p(t, x)$. Let $D = \{D_r, r \geq 0\}$ denote a β -stable subordinator and E_t its first passage time. Then it is known that the density of the time changed processes X_{E_t} is given by $Y(t, x)$ and we have that

$$Y(t, x) = \int_0^\infty p(s, x) f_{E_t}(s) ds,$$

where

$$f_{E_t}(s) = t^{\beta-1} x^{-1-1/\beta} g_\beta \left(tx^{-1/\beta} \right),$$

where $g_\beta(\cdot)$ is the density function of D_1 and is smooth on the real line with $g_\beta(x) = 0$ for $x \leq 0$.

The goal of this paper is to prove the existence and uniqueness of a global solution to (1.1). The work is highly motivated by the recent work by Millet et al [MS21]. It is a well studied phenomena that either a super-linear drift or diffusion term may cause blow-up of the solution. As for the stochastic heat equation, we direct the reader to [FN21; MS93; DKZ19; BG09]. On the other hand, the only other work that we are aware of that is dedicated to proving some non-existence results of (1.1) is [AMN20]. So, to the best of our knowledge, this is the first work on proving the global existence of a solution to (1.1) with a super-linear diffusion term (see Theorem 1.1 below). However, as we will see, the calculations given below are essentially identical to those performed in [MS21] which is the cause for our motivation.

Here is the main result of this chapter. The proof will be postponed until Section 3 below.

C:MainSupLin

Theorem 1.1. *Suppose that σ satisfies (1.2) and the initial data are Borel measurable functions. Then for any $R > 0$ and $(t, x) \in [0, T] \times \bar{B}_R(0)$, where $\bar{B}_R(0)$ denotes the closed ball centered around the origin, there exists a random field solution to (1.1) denoted as $(u(t, x) : (t, x) \in [0, T] \times \bar{B}_R(0))$. This solution is unique and satisfies*

$$\sup_{t \in [0, T], |x| \leq R} |u(t, x)| < \infty, \quad \text{almost surely.} \quad (1.4)$$

The proof of Theorem 1.1 uses a standard stopping time argument along with a series of truncations of the super-linear term, σ , in order to construct a solution of (1.1). Indeed, we define

$$\sigma_N(x) = \begin{cases} \sigma(x) & |x| \leq N \\ \text{continuous and constant} & |x| > N, \end{cases}$$

and then associate for each $N > 0$, the corresponding global solution $u_N(t, x)$, whose existence follows due to Theorem 2.1 below. We then define the stopping time τ_N as follows:

$$\tau_N := \inf \left\{ t > 0 : \sup_{|x| \leq R} |u_N(t, x)| \geq N \right\} \wedge T.$$

We prove that $\{t < \tau_N\} \uparrow \Omega$ and then show that the solution to (1.1) exists pathwise for $\omega \in \Omega$. We remind the reader that in the case of a globally Lipschitz diffusion term, one usually proves that the solution exists as a limit in $L^2(\Omega)$. Because of the stopping-time argument, existence in $L^2(\Omega)$ is no longer guaranteed. Moreover, because $L^2(\Omega)$ existence is no longer guaranteed, then the solution may no longer satisfy the standard Ito isometry.

2 Preliminary results

In this section, we assume that the coefficient σ is globally Lipschitz and therefore satisfies the following inequality:

$$|\sigma(x)| \leq c(\sigma) + L(\sigma)|x|, \quad (2.1)$$

E:SupLinGlo

where $c(\sigma) = \sigma(0)$ and $L(\sigma)$ is the Lipschitz constant of σ and we assume that $L(\sigma) > 0$. The equation (1.1) under the assumption that σ satisfies (2.1) has been studied in [CHN19] where they establish in Theorem 4.1 *ibid* the existence and uniqueness of global random field solutions. In order to do so, one needs to show *Dalang's condition*:

$$\int_0^t ds \int_{\mathbb{R}^d} dy |Y(s, y)|^2 < \infty, \quad \text{for all } t > 0, \quad (2.2)$$

C:Dalang

which is equivalent to (e.g. see Lemma 5.3 *ibid*)

$$d < 2\alpha + \frac{\alpha}{\beta} \min\{2\gamma - 1, 0\} =: \Theta, \quad (2.3)$$

E:DalangEqu

which is further equivalent to

$$\rho(d) > 0 \quad \text{and} \quad d < 2\alpha, \quad (2.4) \quad \boxed{\text{E: DalangEqu}}$$

where

$$\rho(x) := 2\beta + 2\gamma - 1 - \beta x/\alpha. \quad (2.5) \quad \boxed{\text{D: rho}}$$

We now state the existence and uniqueness result proven in [CHN19].

$\boxed{\text{1and2normYZ}}$

Theorem 2.1. [CHN19, Theorem 4.1] Under (2.2), the SPDE (1.1) has a unique (in the sense of versions) random field solution $\{u(t, x) : (t, x) \in (0, \infty) \times \mathbb{R}^d\}$ if the initial data are such that

$$\widehat{C}_T := \sup_{t \in [0, T], x \in \mathbb{R}^d} |J_0(t, x)| < \infty. \quad (2.6) \quad \boxed{\text{E: HomogBdd}}$$

We now state a lemma proven in [CHN19] that calculates the $L^1(\mathbb{R}^d)$ and $L^2(\mathbb{R}^d)$ norms of the fundamental solutions. The lemma will be needed below in some of our calculations.

$\boxed{\text{1and2normYZ}}$

Lemma 2.2. [CHN19, Theorem 4.1 and Lemma 5.5] Assume that $d < 2\alpha$, $\beta \in (0, 2)$, and $\gamma \geq 0$. Then

$$\int_{\mathbb{R}^d} Y(t, x) dx = t^{\beta+\gamma-1} \quad \text{and} \quad \int_{\mathbb{R}^d} Z(t, x) dx = t^{\lceil \beta \rceil - 1}$$

and

$$\int_{\mathbb{R}^d} Y^2(t, x) dx = C_{\#} t^{2(\beta+\gamma-1)-d\beta/\alpha} = C_{\#} t^{\rho-1},$$

for all $t > 0$, where $\rho = \rho(d)$ and

$$C_{\#} := \frac{2}{\Gamma(d/2)(4\pi)^{d/2}} \int_0^\infty u^{d-1} E_{\beta, \beta+\gamma}^2(-u^\alpha) du.$$

Now further assume that $\beta \in (1, 2)$, then

$$\int_{\mathbb{R}^d} Z^*(t, x) dx = 1.$$

Remark 2.3. When $\alpha = \beta = 2$, $\gamma = 0$ and $d = 1$ then Lemma 2.2 gives us that $\int_{\mathbb{R}} Y^2(t, x) dx = t/2$ which coincides with the $L^2(\mathbb{R})$ norm of the wave kernel.

2.1 Some moment bounds

In this section, we will prove some moment bounds and a continuity result for the solution with a globally Lipschitz diffusion term. In the proof of the next proposition, we will use the following two facts:

$$\sup_{t \geq 0} t^k e^{-at} = k^k (ea)^{-k} \text{ for } k \in \mathbb{N} \text{ and } \sup_{t \geq 0} \int_0^t s e^{-as} ds = a^{-2} \text{ for } a > 0. \quad (2.7) \quad \boxed{\text{E: SUPidenti}}$$

$\boxed{\text{P:N(u)Bdd}}$

Proposition 2.4. [MS21, Proposition 3.2] Let u_0 and v_0 be Borel functions satisfying $\|u_0\|_\infty + \|v_0\|_\infty < \infty$. Suppose that $d < 2\alpha$ and $\rho = \rho(d) > 0$, where $\rho(x)$ is defined in (2.5). Then there exists a universal constant $K := K(\alpha, \beta, \gamma, d) > 0$ such that for any

$$a > (8L^2(\sigma)K^2)^{1/\rho}.$$

and

$$p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right].$$

we have that

$$N_{a,p}(u) \leq 2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \quad (2.8) \quad \boxed{\text{E:Nbdd}}$$

where

$$\mathcal{T}_0 = \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

Moreover for any $T > 0$,

$$\sup_{x \in \mathbb{R}^d} \mathbb{E}(|u(t, x)|^p) \leq \exp(apt) \left[2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \right]^p, \quad t \in [0, T]. \quad (2.9) \quad \boxed{\text{E:Pnormtbdd}}$$

Proof. First we fix an $a > 0$ and $p \in [2, \infty)$. Using Lemma 2.2 we see that,

$$\begin{aligned} |J(t, x)| &= \begin{cases} |(Z(t, \cdot) * u_0)(x)| & \beta \in (0, 1] \\ |(Z(t, \cdot) * v_0)(x) + (Z^*(t, \cdot) * u_0)(x)| & \beta \in (1, 2) \end{cases} \\ &\leq \begin{cases} t^{\lceil \beta \rceil - 1} \|u_0\|_\infty & \beta \in (0, 1] \\ t^{\lceil \beta \rceil - 1} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} \end{aligned}$$

Now using (2.7), we see that

$$\begin{aligned} N_{a,p}(J) &\leq \begin{cases} \sup_{t \geq 0} \sup_{x \in \mathbb{R}} t^{\lceil \beta \rceil - 1} \|u_0\|_\infty e^{-at} & \beta \in (0, 1] \\ \sup_{t \geq 0} \sup_{x \in \mathbb{R}} (t^{\lceil \beta \rceil - 1} \|v_0\|_\infty + \|u_0\|_\infty) e^{-at} & \beta \in (1, 2) \end{cases} \\ &= \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases} \quad (2.10) \quad \boxed{\text{E:Jbdd}} \end{aligned}$$

Now we find an upper bound for $N_{a,p}(I)$. By simultaneously applying the Burkholder-Davies-Gundy inequality and triangle inequality, we get that

$$\begin{aligned} \|I(t, x)\|_p^2 &\leq 4p \left\| \int_0^t ds \int_{\mathbb{R}^d} dy Y^2(t - s, x - y) \sigma^2(u(s, y)) \right\|_{p/2} \\ &\leq 4p \int_0^t ds \int_{\mathbb{R}^d} dy Y^2(t - s, x - y) \|\sigma^2(u(s, y))\|_{p/2}. \end{aligned}$$

Now by applying Lemma 2.2 and the Lipschitz property of σ we get that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma)C_\# \int_0^t (t - s)^{\rho - 1} ds + 8pL^2(\sigma)N_{a,p}^2(u) \int_0^t ds \int_{\mathbb{R}^d} dy e^{2as} Y^2(t - s, x - y).$$

Now under Dalang's condition (2.4), the first integral above can be calculated as

$$\int_0^t (t - s)^{\rho - 1} ds = \frac{t^\rho}{\rho}.$$

By applying Lemma 2.2, we can calculate the second integral as

$$\int_0^t ds \int_{\mathbb{R}} dy e^{2as} Y^2(t - s, x - y) = C_\# \int_0^t ds e^{2as} (t - s)^{\rho - 1}.$$

Putting this together gives us that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma)C_\# \frac{t^\rho}{\rho} + 8pL^2(\sigma)N_{a,p}^2(u)C_\# \int_0^t ds e^{2as} (t - s)^{\rho - 1}.$$

Now by multiplying both sides by e^{-2at} , we see that

$$\begin{aligned} \|I(t, x)\|_p^2 e^{-2at} &\leq \frac{8pc^2(\sigma)C_{\#}}{\rho} t^{\rho} e^{-2at} \\ &\quad + 8pL^2(\sigma)N_{a,p}^2(u)C_{\#} \int_0^t ds e^{-2a(t-s)}(t-s)^{\rho-1}. \end{aligned}$$

We now do a change of variables and again recall the definition of the Gamma function to see that

$$\begin{aligned} \|I(t, x)\|_p^2 e^{-2at} &\leq \frac{8pc^2(\sigma)C_{\#}}{\rho} t^{\rho} e^{-2at} + 8pL^2(\sigma)N_{a,p}^2(u)C_{\#} \int_0^{\infty} du e^{-2au}(u)^{\rho-1} \\ &= \frac{8pc^2(\sigma)C_{\#}}{\rho} t^{\rho} e^{-2at} + 8pL^2(\sigma)N_{a,p}^2(u) \frac{C_{\#}\Gamma(\rho)}{(2a)^{\rho}} \end{aligned}$$

where the last line is true if Dalang's condition [CHN19] holds. If we now apply (2.7) and also recall that $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, then we see that

$$N_{a,p}(I) \leq c(\sigma) \sqrt{\frac{8pC_{\#}}{\rho}} \left(\frac{\rho}{2ea}\right)^{\rho/2} + L(\sigma)N_{a,p}(u) \sqrt{\frac{8pC_{\#}\Gamma(\rho)}{(2a)^{\rho}}}. \quad (2.11) \quad \boxed{\text{E:SoleMil3.}}$$

To simplify things, we now let

$$K := \max \left\{ \sqrt{\frac{8C_{\#}}{\rho}} \left(\frac{\rho}{2e}\right)^{\rho/2}, \sqrt{\frac{8C_{\#}\Gamma(\rho)}{2^{\rho}}} \right\}$$

then,

$$N_{a,p}(I) \leq \frac{K\sqrt{p}}{a^{\rho/2}} (c(\sigma) + L(\sigma)N_{a,p}(u)). \quad (2.12) \quad \boxed{\text{E:Ibdd}}$$

Now by combining (2.10) and (2.12), we see that

$$N_{a,p}(u) \leq \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_{\infty} & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_{\infty} + \|u_0\|_{\infty} & \beta \in (1, 2). \end{cases} \quad (2.13)$$

$$+ K \frac{\sqrt{p}c(\sigma)}{a^{\rho/2}} + K \frac{\sqrt{p}L(\sigma)}{a^{\rho/2}} N_{a,p}(u). \quad (2.14) \quad \boxed{\text{E:Nbddgen}}$$

It is important to note that if we now restrict ourselves to the case of $\alpha = \beta = 2$, $d = 1$ and $\gamma = 0$, then (2.11) and (2.14) recover [MS21, (3.9) and (3.10)] respectively. Now choose $a > 0$ such that

$$a > (8L^2(\sigma)K^2)^{1/\rho}.$$

This implies that the following interval is nonempty:

$$\left[2, \frac{a^{\rho}}{4L^2(\sigma)K^2} \right].$$

In addition, for any p in this interval, we have that

$$\frac{\sqrt{p}L(\sigma)K}{a^{\rho/2}} \leq \frac{1}{2},$$

and we also have

$$\frac{\sqrt{p}}{a^{\rho/2}} \leq \frac{1}{2KL(\sigma)}.$$

This along with (2.14) implies that

$$N_{a,p}(u) \leq 2 \begin{cases} ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|u_0\|_\infty & \beta \in (0, 1] \\ ([\beta] - 1)^{[\beta]-1} (ea)^{1-[\beta]} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} + \frac{c(\sigma)}{L(\sigma)}$$

and this is precisely (2.8). Note that (2.9) follows immediately from (2.8). $\square$

We will need to recall the following result proved in [CHN19] in order to prove Proposition 2.7 below.

P:YInc

Proposition 2.5. [CH21, Proposition 2.1 and 2.2] *Suppose that $\alpha \in (0, 2]$, $\beta \in (0, 2)$, $\gamma > 0$, and (2.3) holds. Consider $\rho = \rho(d)$ which is defined above in (2.5). Then $Y(t, x) = Y_{\alpha, \beta, \gamma, d}(t, x)$ satisfies the following:*

P:YIncCase1

1. *Suppose that $0 < \theta < (\Theta - d) \wedge 2$ where Θ is given in (2.3), $0 < t, r \leq T$ for some $T > 0$, $\beta \leq 1$ and $\gamma \leq [\beta] - \beta$. Then there exists some $C := C(\alpha, \beta, \gamma, \nu, d, \theta, T)$ such that*

$$\int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \leq C \left(|t-r|^\rho + |x-z|^\theta \right).$$

P:YIncCase2

2. *If we only assume that $\alpha \in (0, 2]$, $\beta \in (0, 2)$, $\gamma > 0$, $0 < t, r \leq T$ for some $T > 0$ and (2.3) holds then for $0 < \theta < (\Theta - d) \wedge 2$ there exists some $C := C(\alpha, \beta, \gamma, \nu, d, \theta, T)$ such that*

$$\int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \leq C \left(|t-r|^q + |x-z|^\theta \right),$$

where

$$0 < q < \rho.$$

Remark 2.6. In Proposition 2.5 above, we may often ignore the dependence of C on $\alpha, \beta, \gamma, \nu, d$ and just write $C(\theta, T)$.

P:NormInc

Proposition 2.7. *Let $a > 0$ and suppose that $0 < \theta < (\Theta - d) \wedge 2$ and $\rho = \rho(d) > 0$. Then for any $x, z \in \mathbb{R}$, $p \geq 2$ and for any $T > 0$ with $t \in [0, T]$, the stochastic integral, $I(t, x)$, satisfies the following:*

$$\frac{\|I(t, x) - I(r, z)\|_p}{(|t-r|^q + |x-z|^\theta)^{1/2}} \leq C(p, \theta, T) [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)], \quad (2.15)$$

E:NormInc_N

where

$$\mathcal{M}_1 = \sqrt{p}c(\sigma) \quad \text{and} \quad \mathcal{M}_2 = \sqrt{p}L(\sigma),$$

and

1. $0 < q \leq \rho$ under Case 1 of Proposition 2.5,
2. $0 < q < \rho$ under Case 2 of Proposition 2.5.

Moreover, for any

$$a > (8L^2(\sigma)K^2)^{1/\rho}.$$

and

$$p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right]$$

we have that

$$\frac{\|I(t, x) - I(r, z)\|_p}{(|t-r|^q + |x-z|^\theta)^{1/2}} \leq C(p, \theta, T) \left[\mathcal{M}_1 + \mathcal{M}_2 e^{aT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right) \right]. \quad (2.16)$$

E:NormInc_c

Proof. Note that because of the initial conditions, we see that $\|u(t, x) - u(r, z)\|_p = \|I(t, x) - I(r, z)\|_p$. We now apply Burkholder-Davies-Gundy inequality along with the triangle inequality, as was done in the proof of Proposition 2.4, to see that

$$\begin{aligned} \|I(t, x) - I(r, z)\|_p^2 &\leq 4p \left\| \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \sigma^2(u(s, y)) \right\|_{p/2} \\ &\leq 4p \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \|\sigma^2(u(s, y))\|_{p/2}. \end{aligned}$$

Now by applying the Lipchitz condition on σ gives us that

$$\begin{aligned} \|I(t, x) - I(r, z)\|_p^2 &\leq 8p \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \left(c^2(\sigma) + L^2(\sigma) \|u(s, y)\|_p^2 \right) \\ &\leq 8p \left[c^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right. \\ &\quad \left. + L^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \exp(2as) N_{a,p}^2(u) \right] \\ &\leq 8p \left[c^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right. \\ &\quad \left. + L^2(\sigma) \exp(2aT) N_{a,p}^2(u) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right]. \end{aligned}$$

We now apply Proposition 2.5 to see that

$$\begin{aligned} \|I(t, x) - I(r, z)\|_p^2 &\leq 8p \left[c^2(\sigma) C(\theta, T) \left(|t-r|^q + |x-z|^\theta \right) \right. \\ &\quad \left. + L^2(\sigma) \exp(2aT) N_{a,p}^2(u) C(\theta, T) \left(|t-r|^q + |x-z|^\theta \right) \right], \end{aligned}$$

where $C(\theta, T)$ is as in Proposition 2.5. Now by taking the square root of both sides and applying the identity $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ yields

$$\|I(t, x) - I(r, z)\|_p \leq 2\sqrt{2C(\theta, T)p} \left[c(\sigma) + L(\sigma) \exp(aT) N_{a,p}(u) \right] \left(|t-r|^q + |x-z|^\theta \right)^{1/2}.$$

This proves (2.15). It is clear that (2.16) follows directly from (2.15) and Proposition 2.4. $\square$

P:Holder

Proposition 2.8. *The stochastic integral, I , has a version, still denoted by I that is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$ where $0 < \theta < (\Theta - d) \wedge 2$ and*

1. $0 < q \leq \rho$ under Case 1 of Proposition 2.5,
2. $0 < q < \rho$ under Case 2 of Proposition 2.5.

Moreover, if we consider

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right],$$

where K is the constant from Proposition 2.4, then

$$\mathbb{E} \left[\sup_{t \in [0, T], |x| \leq R} |u(t, x)|^p \right] \leq 2^{p-1} \widehat{C}_T^p + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right)^p \right],$$

where $\widehat{C}_T$ is defined in (2.6).

Proof. Proposition 2.7 implies that

$$\|I(t, x) - I(r, z)\|_p \leq C(p, \theta, T) [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)] \left(|t - r|^q + |x - z|^\theta \right)^{1/2}. \quad (2.17)$$

By Kolmogorov's continuity theorem, there exists a version of I , which we denote by I , that is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$.

Next, note that by triangle inequality,

$$\begin{aligned} |u(t, x)|^p &\leq 2^{p-1} (\|J_0(\cdot, \circ)\|_\infty^p + |u(t, x) - J_0(t, x)|^p) \\ &\leq 2^{p-1} \left(\widehat{C}_T^p + \sup_{t \in [0, T], |x| \leq R} |I(t, x)|^p \right) \\ &= 2^{p-1} \left(\widehat{C}_T^p + \sup_{t \in [0, T], |x| \leq R} |I(t, x)|^p \right), \end{aligned}$$

where we recall that $\widehat{C}_T$ defined in (2.6). Due to the continuity of I on the compact set $[0, T] \times \bar{B}_R(0)$, there exists a point $(t_0, x_0) \in [0, T] \times \bar{B}_R(0)$ such that

$$\sup_{t \in [0, T], |x| \leq R} |I(t, x)|^p = |I(t_0, x_0)|^p.$$

Thus, for any $(t, x) \in [0, T] \times \bar{B}_R(0)$,

$$|u(t, x)|^p \leq 2^{p-1} \left(\widehat{C}_T^p + |I(t_0, x_0)|^p \right)$$

which in return implies that

$$\sup_{t \in [0, T], |x| \leq R} |u(t, x)|^p \leq 2^{p-1} \left(\widehat{C}_T^p + |I(t_0, x_0)|^p \right).$$

Now by taking the expectation of both sides and applying Proposition 2.7 gives us

$$\begin{aligned} \mathbb{E} \left[\sup_{t \in [0, T], |x| \leq R} |u(t, x)|^p \right] &\leq 2^{p-1} \widehat{C}_T^p + 2^{p-1} C(p, \theta, T)^p \left((2T)^{q/2} + (2R)^{\theta/2} \right)^p [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)]^p \\ &= 2^{p-1} \widehat{C}_T^p + C(p, \theta, T, R) [\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} N_{a,p}(u)^p]. \end{aligned}$$

Now if we consider

$$a > (8L^2(\sigma)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma)K^2} \right],$$

then by Proposition 2.4, we have that

$$\mathbb{E} \left[\sup_{t \in [0, T], |x| \leq R} |u(t, x)|^p \right] \leq 2^{p-1} \widehat{C}_T^p + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right)^p \right].$$

□

Remark 2.9. The Hölder continuity in time proven above in Proposition 2.8 recovers the result proven by Foondun and Nane in [FN17, Theorem 1.9] if we restrict ourself to the case where $\beta \in (0, 1)$ and $\gamma = 1 - \beta$.

3 Proof of Theorem 1.1

Proof. We first solve a *truncated* version of (1.1). By this, we mean (1.1) with σ replaced by σ_N where in the case of $d = 1$,

$$\sigma_N(x) = \sigma(x)\mathbf{1}_{\{|x| \leq N\}} + \sigma(N)\mathbf{1}_{\{x > N\}} + \sigma(-N)\mathbf{1}_{\{x \leq -N\}},$$

and in higher dimensions σ_N is defined similarly. Note that since σ satisfies (1.2), then σ_N is Lipschitz continuous with Lipschitz constant $L(\sigma_N) := \sigma_2 \ln(2N)^\delta$. In other words,

$$|\sigma_N(x)| \leq \sigma_1 + \sigma_2 \ln(2N)^\delta |x|.$$

Hence we can apply Theorem 2.1 and see that there exists a unique solution of the truncated version of (1.1), which we denote by $u_N := \{u_N(t, x) : (t, x) \in [0, T] \times \mathbb{R}\}$. In addition, Proposition 2.8 implies that the solution has a version, still denoted by u_N , which is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$.

We will now apply Proposition 2.8 to find an upper bound on the p -norm of u_N . For this, consider

$$a > (8L^2(\sigma_N)K^2)^{1/\rho} \quad \text{and} \quad p \in \left[2, \frac{a^\rho}{4L^2(\sigma_N)K^2}\right].$$

Then Proposition 2.8 implies that

$$\mathbb{E} \left(\sup_{t \in [0, T], x \in [-R, R]} |u_N(t, x)|^p \right) \leq 2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p(N) \exp(apT) \left(2 + \frac{\sigma_1}{L(\sigma_N)} \right) \right] \quad (3.1)$$

where we recall that

$$\mathcal{M}_1 = \sqrt{p}\sigma_1 \quad \text{and} \quad \mathcal{M}_2(N) = \sqrt{pL(\sigma_N)} = \sqrt{p\sigma_2 \ln(2N)^\delta}. \quad (3.2)$$

We now use the above to prove the existence and uniqueness of (1.1) with superlinear σ satisfying (1.2). For any $T > 0$ and $N \geq 2$ with $N \in \mathbb{N}$, we define the following stopping time:

$$\tau_N := \inf \left\{ t > 0 : \sup_{|x| \leq R} |u_N(t, x)| \geq N \right\} \wedge T. \quad (3.3)$$

The uniqueness of the solution and the local property of the stochastic integral imply that on $\{t < \tau_N\}$, $u_N(t, x) = u_{N+k}(t, x)$ for any $k \in \mathbb{N}$. Hence, $(\tau_N)_{N \geq 2}$ is increasing and bounded above by T .

We now momentarily assume that $\sup_N \tau_N = T$ which would imply that $\{t < \tau_N\} \uparrow \Omega$. On each $\{t < \tau_N\}$, define $(u(t, x) : (t, x) \in [0, T] \times \mathbb{R})$ by $u(t, x) = u_N(t, x)$ and hence $u(t, x) = u_{N+k}(t, x)$ for any $k \in \mathbb{N}$. This implies that on $\{t < \tau_N\}$

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma_N(u(s, y)) W(ds, dy).$$

However, on $\{t < \tau_N\}$, we have that $\sigma_N(u_N(t, x)) = \sigma(u(t, x))$ and so u satisfies

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad \{t < \tau_N\}.$$

Lastly, since $\{t < \tau_N\} \uparrow \Omega$, we conclude that

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad (t, x) \in [0, T] \times \mathbb{R}. \quad (3.4)$$

It is mentioned that the stochastic integrals are no longer defined in $L^2(\Omega)$. The extension of the integral is found in Dalang and Sanz-Sole's textbook that is not yet published.

We now focus on proving that $\sup_N \tau_N = T$, or equivalently that $\mathbb{P}(\tau_N < T) \rightarrow 0$ as $N \rightarrow \infty$. To do this, note that by Chebychev's inequality

$$\begin{aligned} \mathbb{P}(\tau_N < T) &\leq \mathbb{P}\left(\sup_{t \in [0, T], |x| \leq R} |u_N(t, x)| \geq N\right) \\ &\leq N^{-p} \mathbb{E}\left(\sup_{t \in [0, T], |x| \leq R} |u_N(t, x)|^p\right). \end{aligned}$$

An application of (3.1) gives

$$\begin{aligned} N^{-p} \mathbb{E}\left(\sup_{t \in [0, T], |x| \leq R} |u_N(t, x)|^p\right) &\leq N^{-p} \left(2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p\right]\right) \\ &= \frac{2^{p-1} + C(p, \theta, T, R) \mathcal{M}_1^p}{N^p} + \frac{C(p, T, R) \mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p}{N^p}. \end{aligned}$$

The first term clearly converges to 0 as $N \rightarrow \infty$. As for the second term, we expand using (3.2) to see that

$$\begin{aligned} \frac{\mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p}{N^p} &= \frac{\left(\sqrt{p\sigma_2 \ln(2N)^\delta}\right)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{\sqrt{p\sigma_2 \ln(2N)^\delta}}\right)^p}{N^p} \\ &\sim \frac{\ln(2N)^{p\delta/2}}{N^p} \rightarrow 0, \end{aligned}$$

where above we use the symbol $\sim$ to mean asymptotically equivalent as $N \rightarrow \infty$. This completes the proof of the Theorem. $\square$

4 Some comments on a spatially colored noise

Consider equation 1.1 driven by a noise that is white in time and colored space. In other words, W is a 0-mean Gaussian process indexed on the space of Schwartz functions with the following covariance structure:

$$C(\phi, \psi) := \int_0^\infty dt \int_{\mathbb{R}^d} \Gamma(dx) (\phi(t, \cdot) * \tilde{\psi}(t, \cdot))(x), \quad \tilde{\psi}(x) = \psi(-x). \quad (4.1) \quad \boxed{\text{E:Colored}}$$

As per usual, the tempered measure Γ is non-negative and nonnegative definite and satisfies the following two conditions:

NoiseAssum1

Assumption 4.1.

1. the spectral measure μ is such that $\int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{1+|\zeta|^\alpha} < \infty$,
2. Γ is absolutely continuous with respect to the Lebesgue measure and therefore has a density, $\Gamma(dx) = f(x)dx$. Furthermore, when this is the case, we may often write $\mu(d\zeta) = \varphi(\zeta)d\zeta$ where we have the relation $\varphi = \mathcal{F}f$,

Under this assumption, we may rewrite (4.1) as

$$C(\phi, \psi) = \int_0^\infty dt \int_{\mathbb{R}^{2d}} dx dy \phi(t, x) f(x - y) \psi(t, y)$$

and by applying Placherel's identity, this further implies that

$$\int_0^\infty dt \int_{\mathbb{R}^{2d}} dx dy \phi(x) f(x-y) \psi(y) = (2\pi)^{-d} \int_0^\infty dt \int_{\mathbb{R}^{2d}} \mu(d\zeta) \mathcal{F}(\phi(t, \cdot))(\zeta) \overline{\mathcal{F}(\psi(t, \cdot))(\zeta)}.$$

We now prove a lemma which is a generalization of equation (2.2) which accounts for the colored noise.

L:YfYbdd

Lemma 4.2. *Suppose Assumption 4.1 holds. Then there exists a constant $C(\beta, \gamma) =: C > 0$ such that for any $x, y \in \mathbb{R}^d$ we have that*

$$\int_{\mathbb{R}^{2d}} Y(t, x-z) Y(t, y-w) f(w-z) dz dw \leq C \left(\frac{\nu}{2}\right)^{-\lambda/\alpha} t^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1+|\xi|^\alpha)^2} < \infty \quad (4.2)$$

E:YYfBdd

Proof. Through a change of variables, we may write the left side of (4.2) as the following:

$$\int_{\mathbb{R}^{2d}} Y(t, x-z) Y(t, y-w) f(w-z) dz dw = \int_{\mathbb{R}^d} Y(t, u - (x-y)) (Y(t, \cdot) * f)(u) du,$$

where $*$ denotes the spatial convolution. Now an application of Parseval's identity gives us that

$$\int_{\mathbb{R}^d} Y(t, u - (x-y)) (Y(t, \cdot) * f)(u) du = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathcal{F}Y(t, \cdot - (x-y))(\zeta) \mathcal{F}(Y(t, \cdot) * f)(\zeta) d\zeta.$$

Applying the translation property of the Fourier transform and [CHN19, Theorem 4.1] gives us that

$$\begin{aligned} \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathcal{F}Y(t, \cdot - (x-y))(\zeta) \mathcal{F}(Y(t, \cdot) * f)(\zeta) d\zeta \\ = \frac{t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} \exp(-i(x-y)\zeta) E_{\beta, \beta+\gamma} \left(-\frac{\nu t^\beta |\zeta|^\alpha}{2} \right)^2 \mu(d\zeta). \end{aligned}$$

Hence by taking the absolute value on both sides and recalling the positivity of f , Y and the Mittag-Leffler function we get that

$$\int_{\mathbb{R}^{2d}} Y(t, x-z) Y(t, y-w) f(w-z) dz dw \leq \frac{t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} E_{\beta, \beta+\gamma} \left(-\frac{\nu t^\beta |\zeta|^\alpha}{2} \right)^2 \mu(d\zeta).$$

Now, by applying [Pod99, Theorem 1.6], we see that there exists a $C > 0$ depending only on β and γ such that

$$\frac{t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} E_{\beta, \beta+\gamma} \left(-\frac{\nu t^\beta |\xi|^\alpha}{2} \right)^2 \mu(d\zeta) \leq \frac{C t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1 + \frac{\nu}{2} t^\beta |\zeta|^\alpha)^2}.$$

Another change of variables implies that

$$\frac{C t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1 + \frac{\nu}{2} t^\beta |\zeta|^\alpha)^2} \leq \frac{C}{(2\pi)^d} \left(\frac{\nu}{2}\right)^{-\lambda/\alpha} t^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_{\mathbb{R}^d} \frac{\mu(d\zeta)}{(1 + |\xi|^\alpha)^2}.$$

The finiteness of the above integral follows from Assumption 4.1. $\square$

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